

DANIELE SALA

- » Status: M.Sc. in Environmental and Land Planning Engineering
- » Focus: Process-based Hydrological Modelling, Drought and Hydro-meteorological Monitoring, Satellite Data Integration for Environmental Early Warning
- » Skills: Python, R, MATLAB, QGIS, Remote Sensing (multispectral, UAV/drone imagery), GPS-based spatial surveys, Time-Series Analysis, GIS-based morphological mapping, Fortran, SQL
- » Languages: Italian (Native), English (C1), Spanish (B2)



»»» Education

2021 - 2024	M.Sc in Environmental and Land Planning Engineering Final Grade: 106/110 Weighted Average: 27.31/30 <i>Relevant coursework:</i> » Hydrogeological Risk (30/30) · Remote Sensing (30/30) · Statistical Models and Stochastic Processes (29/30) · Ecosystems Conservation and Management (29/30) Master Thesis: Impact of Agricultural Irrigation on the Hydrologic Regime of the Po River Basin Supervisor: Prof. Giovanni Ravazzani, DICA. <i>Research objectives and methodology:</i> » Performed catchment-scale water balance and water accounting analysis, quantifying irrigation impacts on the hydrological cycle » Developed and calibrated the FEST hydrological model to simulate land surface and subsurface water dynamics across the Po River Basin, integrating 10 years of meteorological, hydrological, and land-use spatial data within a GIS framework » Conducted comprehensive sensitivity and uncertainty analysis on key model parameters and developed counterfactual scenarios contrasting current irrigation practices, alternative withdrawal strategies and drought conditions, isolating the contribution of human water-use decisions from purely climatic drivers <i>Key findings:</i> » Quantified 15% reduction in river flows during irrigation season (April-September), with altered evapotranspiration patterns and decreased aquifer levels throughout the year » Simulated and evaluated drought-stress scenarios by systematically varying irrigation withdrawal intensity, providing quantitative insight into soil moisture depletion dynamics and groundwater recharge deficits under water-limited conditions. <i>Field work and stakeholder engagement:</i> » Extensive collaboration with regional irrigation consortia for data collection, characterization of major water diversions, and model validation <i>Technical skills applied:</i> » Process-based numerical modelling (FEST, Fortran), GIS-based spatial analysis, thematic and hazard mapping, environmental time-series processing, scenario development under climate stress conditions	Politecnico di Milano
2022 - 2023	Erasmus Exchange Semester » Specialization in Water Quality, Pollution Transport in Natural Environments, and Wastewater Treatment Management	UPV, Valencia, Spain
2018 - 2021	B.Sc in Environmental Engineering » Foundation in environmental engineering principles, fluid mechanics, hydrology, and environmental systems	Politecnico di Milano

Professional Experience

2025 - Present	Natural Risk Assessment Specialist	Progesi S.p.a., Milan
<i>Main responsibilities:</i>		
» Develop and maintain integrated environmental monitoring systems processing multi-source data from ground stations, satellite platforms, and remote sensing imagery to support proactive natural hazard management and risk quantification at regional scale		
» Operational use of ICON-I2 and ECMWF-IFS NWP model outputs as meteorological forcing for regional hydrological forecasting, including calibration of precipitation and evapotranspiration estimates against ground-station and satellite observations over the Po Plain and Alpine foothills.		
» Process complex environmental datasets from ground sensors, satellite imagery, and remote sensing products using spatial analysis and GIS techniques; produce technical reports, interactive dashboards, and data visualizations to disseminate environmental monitoring results to institutional and scientific stakeholders		
» Process complex environmental datasets from multiple sources (meteorological stations, hydrological sensors, satellite imagery) using spatial analysis techniques, and produce technical reports, interactive dashboards, and data visualization tools to communicate environmental complexity to institutional stakeholders		
» Collaborate with university research groups and PhD teams on applied hydrometeorological and climate modelling projects, contributing to peer-reviewed research outputs and bridging operational forecasting with scientific investigation of climate-driven extremes		
<i>Selected impact case:</i>		
» September 2025 Seveso & Bevera flood event: Led end-to-end flood risk management workflow from hydrological alerting and real-time monitoring to post-event impact assessment and institutional reporting, supporting municipalities and Regione Lombardia in emergency state declarations and public infrastructure damage compensation claims		

2024	Climate Data Analyst	Enel S.p.a., Milan
<i>Main responsibilities:</i>		
» Analyzed climate-related datasets to assess physical and transition risks for a global energy infrastructure portfolio, developing analytical workflows and automation tools using Python and R		
<i>Key contribution:</i>		
» CSRD Alignment Project: Conducted cross-indicator sustainability benchmarking analysis, supporting alignment between climate performance metrics and regulatory reporting frameworks		

References

Prof. Giovanni Ravazzani	Politecnico di Milano
» Thesis Supervisor · Department of Civil and Environmental Engineering (DICA)	
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Prof. Marco Gianinetto	Politecnico di Milano
» Remote Sensing · Department of Civil and Environmental Engineering (DICA)	
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Prof. Ilenia Epifani	Politecnico di Milano
» Statistical Models and Stochastic Processes · Department of Mathematics (DMAT)	
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